

HARSHITH MULAKALA

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EDUCATION

The University of Texas at Dallas Bachelor of Science | *Computer Science*

May 2028

Honors: Academic Excellence Scholarship

WORK EXPERIENCE

Majestic Turbo - *AI Engineer Intern*; Dallas, TX

July 2026 - August 2026

Internal AI tooling and sales automation systems for a turbocharger company

- Built five internal AI tools for a turbocharger company, including an inventory management system and an executive command center for delegating tasks to sales reps and tracking shipments.
- Engineered a late-shipment prediction feature and a call-recording pipeline that transcribes sales calls, summarizes them with LLMs, and writes summaries into each rep's CRM record.
- Developed a marketing analytics dashboard and a lead-generation tool using the Exa search API and Google Places API to surface recent vehicle crashes and nearby repair shops for outreach.

Basics - *Founder & CTO*; San Francisco, CA

January 2026 - July 2026

Self-hosted company operating system replacing fragmented SaaS tools with a unified workspace for early-stage startups

- Architected a self-hosted startup operating system in Next.js, TypeScript, Node.js, PostgreSQL, Docker, and AWS, unifying CRM, projects, documents, analytics, and automations; developed in the open with a four-engineer contributor team at github.com/Basics-Vault/basicsOS.
- Designed multi-tenant RBAC, event-driven workflows, real-time data synchronization, extensible API integrations, and containerized deployment for secure customer-managed environments.
- Converted insights from 50+ founder interviews into a three-pillar technical roadmap; launched four paid pilots in 20 days and raised a \$1M pre-seed at a \$10M valuation.

Speakl Inc. - *Co-Founder & CTO*; Frisco, TX

June 2025 - December 2025

AI communication platform delivering multimodal feedback for public speaking and interview performance

- Architected a React and FastAPI platform serving 1.5K lifetime users and 300+ monthly active users through custom computer vision, audio, and NLP pipelines.
- Engineered AWS infrastructure processing 15K+ monthly API requests at 99% uptime using S3, DynamoDB, IAM, auto-scaling, and CloudWatch monitoring.
- Led four engineers, implemented CI/CD and code-review standards, and shipped 50+ production deployments while processing 500+ hours of speech across 10+ feedback metrics.

ACM Research - *Head Researcher*; Richardson, TX

September 2024 - December 2024

AdFusion research project applying generative AI to automated advertisement creation

- Fine-tuned Stable Diffusion and large language models on 5,000+ ad images using preprocessing, augmentation, and output-validation pipelines.
- Built Python and Selenium scrapers collecting 3,000+ ads from 10+ platforms with 95%+ consistency through deduplication and robust error handling.

iTallentii Tech - *Software Engineering Intern & Consultant*; Frisco, TX

November 2022 - January 2023

Software engineering internship and follow-on consulting across data collection, API integration, and full-stack delivery

- Developed JavaScript and Node.js web-scraping pipelines across 20+ public sources, reducing manual data collection time by 80%.
- Integrated a React front end with RESTful Node.js APIs and fault-tolerant error handling; returned as a consultant to build the katars.com website and supporting internal tools.

PROJECTS

PlayAI | C#, Python, Unity

GitHub

- Built an AI application that converts a single text prompt into a playable Unity 2D game, generating C# gameplay scripts, textures, and scene assets automatically.
- Orchestrated a C# Unity runtime alongside Python generation pipelines that validate, compile, and load model-authored code into an executable game build.

ClipThat! | Swift, Xcode

App Store

- Shipped a privacy-first iOS app live on the App Store that continuously buffers on-device audio and saves the prior 30-60 seconds with one tap and zero cloud storage.
- Designed a rolling native audio buffer using background capture and local persistence so users recover recent moments without any continuous cloud recording.

HeyMom - Real-Time Voice Agent | React Native, Python, FastAPI, WebSockets, Deepgram

GitHub

- Built a real-time voice agent that streams press-to-talk microphone audio from a React Native (Expo) client over a WebSocket connection to a FastAPI backend for live transcription.
- Engineered a low-latency AI voice pipeline with Deepgram speech-to-text and Aura 2 text-to-speech, streaming synthesized audio chunks back to the device for immediate playback.

Q Bot | Node.js, JavaScript, Discord.js

GitHub

- Built a voice-controlled Discord bot that recognizes spoken commands in live voice channels to search, queue, and stream music with real-time audio processing.
- Implemented an event-driven Node.js command architecture with queue state management and fault-tolerant handlers for reliable, low-latency interactions across concurrent servers.

ADDITIONAL INFORMATION

Languages: Python, Java, JavaScript/TypeScript, C#, Swift, C/C++, SQL

Frameworks & Cloud: React, React Native, Next.js, Node.js, FastAPI, Flask, PyTorch, Unity, Selenium, AWS, Docker, PostgreSQL, Supabase

Tools & Certifications: Git, CI/CD, REST APIs, WebSockets, Deepgram, Exa Search API, Google Places API, Jira; Certiport Certified in Java, JavaScript, and Network Security